

Quick-Bite Relational Schema

Formal Boyce-Codd Normal Form (BCNF) Proof & Verification

1. Introduction & BCNF Definition

This document provides a formal proof that the relational schema derived from the **Quick-Bite** Entity-Relationship (ER) diagram is in **Boyce-Codd Normal Form (BCNF)**.

Definition: A relational schema R is in BCNF if and only if, for every one of its non-trivial functional dependencies (FDs) $X \rightarrow Y$, X is a superkey. In simpler terms, every determinant must be a candidate key.

2. Table-by-Table BCNF Proof

1. Users

Users(User_ID, First_Name, Last_Name, Email, Phone, Password_Hash, Created_At)

- **Candidate Keys:** *User_ID, Email, Phone*
- **Functional Dependencies:**
 - *User_ID → First_Name, Last_Name, Email, Phone, Password_Hash, Created_At*
 - *Email → User_ID, First_Name, Last_Name, Phone, Password_Hash, Created_At*
 - *Phone → User_ID, First_Name, Last_Name, Email, Password_Hash, Created_At*

Proof: All determinants (*User_ID, Email, Phone*) are candidate keys. Thus, the table is in BCNF.

2. Addresses

Addresses(Address_ID, *User_ID*, Add_Line_1, Add_Line_2, Area, City, Pincode)

- **Candidate Keys:** *Address_ID*
- **Functional Dependencies:**
 - *Address_ID* → *User_ID, Add_Line_1, Add_Line_2, Area, City, Pincode*

Proof: The only determinant is *Address_ID*, which is the Primary Key. Thus, it is in BCNF.

3. Wallets

Wallets(Wallet_ID, *User_ID*, Balance, Created_At, Updated_At)

- **Candidate Keys:** *Wallet_ID, User_ID* (Assuming 1-to-1 relationship between User and Wallet)
- **Functional Dependencies:**
 - *Wallet_ID* → *User_ID, Balance, Created_At, Updated_At*
 - *User_ID* → *Wallet_ID, Balance, Created_At, Updated_At*

Proof: All determinants (*Wallet_ID, User_ID*) are candidate keys. Thus, it is in BCNF.

4. Wallet_Transactions

Wallet_Transactions(Transaction_ID, *Wallet_ID*, Amount, Type, Status, Created_At)

- **Candidate Keys:** *Transaction_ID*
- **Functional Dependencies:**
 - *Transaction_ID* → *Wallet_ID, Amount, Type, Status, Created_At*

Proof: The only determinant is *Transaction_ID*, which is a superkey. BCNF satisfied.

5. Restaurants

Restaurants(Restaurant_ID, Name, Phone_No, Email, FSSAI_License, Open_Time, Close_Time, Address_ID, Total_Rating)

- **Candidate Keys:** *Restaurant_ID, FSSAI_License, Phone_No, Email*
- **Functional Dependencies:**
 - *Restaurant_ID → Name, Phone_No, Email, FSSAI_License, Open_Time, Close_Time, Address_ID, Total_Rating*
 - *FSSAI_License → Restaurant_ID, Name, ...*

Proof: All non-trivial determinants are candidate keys. BCNF satisfied.

6. Cuisines & 7. Restaurant_Cuisines

Cuisines(Cuisine_ID, Name, Description)

Restaurant_Cuisines(Restaurant_ID, Cuisine_ID)

- **Candidate Keys:** *Cuisine_ID* (Cuisines), *{Restaurant_ID, Cuisine_ID}* (Mapping Table)
- **Functional Dependencies:**
 - *Cuisine_ID → Name, Description*
 - *{Restaurant_ID, Cuisine_ID} → ∅* (All attributes form the PK)

Proof: Determinants are PKs. Both tables are in BCNF.

8. Menu_Categories & 9. Menu_Items

Menu_Categories(Category_ID, Restaurant_ID, Name)

Menu_Items(Item_ID, Category_ID, Name, Description, Price, Is_Veg, Is_Available)

- **Candidate Keys:** *Category_ID, Item_ID*
- **Functional Dependencies:**
 - *Category_ID → Restaurant_ID, Name*
 - *Item_ID → Category_ID, Name, Description, Price, Is_Veg, Is_Available*

Proof: In both tables, the only determinant is the primary key. BCNF satisfied.

10. Carts & 11. Cart_Items

Carts(Cart_ID, User_ID, Total_Amount, Created_At)

Cart_Items(Cart_ID, Item_ID, Quantity)

- **Candidate Keys:** *Cart_ID* (Carts), *{Cart_ID, Item_ID}* (Cart_Items)
- **Functional Dependencies:**
 - *Cart_ID → User_ID, Total_Amount, Created_At*
 - *{Cart_ID, Item_ID} → Quantity*

Proof: Determinants are strictly the superkeys of their respective relations. BCNF satisfied.

12. Orders & 13. Order_Items

Orders(Order_ID, *User_ID*, *Restaurant_ID*, *Cart_ID*, Total_Amount, Status, Created_At)

Order_Items(Order_ID, Item_ID, Quantity, Price_At_Order)

- **Candidate Keys:** *Order_ID*, {*Order_ID*, *Item_ID*}
- **Functional Dependencies:**
 - *Order_ID* → *User_ID*, *Restaurant_ID*, *Cart_ID*, *Total_Amount*, *Status*, *Created_At*
 - {*Order_ID*, *Item_ID*} → *Quantity*, *Price_At_Order*

Proof: Every determinant is a candidate key. BCNF satisfied.

14. Delivery_Partners & 15. Deliveries

Delivery_Partners(Partner_ID, Name, Phone_No, Vehicle_No, Rating)

Deliveries(Delivery_ID, *Order_ID*, *Partner_ID*, Status, Pickup_Time, Delivery_Time)

- **Candidate Keys:** *Partner_ID*, *Delivery_ID*, *Order_ID* (assuming 1 delivery per order)
- **Functional Dependencies:**
 - *Partner_ID* → *Name*, *Phone_No*, *Vehicle_No*, *Rating*
 - *Delivery_ID* → *Order_ID*, *Partner_ID*, *Status*, *Pickup_Time*, *Delivery_Time*
 - *Order_ID* → *Delivery_ID*, *Partner_ID*, *Status*, *Pickup_Time*, *Delivery_Time*

Proof: All determinants are candidate keys. BCNF satisfied.

16. Payments

Payments(Payment_ID, *Order_ID*, Mode, Amount, Transaction_Ref, Status, Created_At)

- **Candidate Keys:** *Payment_ID*, *Transaction_Ref*
- **Functional Dependencies:**
 - *Payment_ID* → *Order_ID*, *Mode*, *Amount*, *Transaction_Ref*, *Status*, *Created_At*
 - *Transaction_Ref* → *Payment_ID*, *Order_ID*, *Mode*, *Amount*, *Status*, *Created_At*

Proof: Determinants are candidate keys. BCNF satisfied.

17. Reviews & 18. Complaints

Reviews(Review_ID, *Order_ID*, *User_ID*, *Restaurant_ID*, Rating, Comment, Created_At)

Complaints(Complaint_ID, *Order_ID*, *User_ID*, Description, Status, Created_At)

- **Candidate Keys:** *Review_ID*, *Complaint_ID*
- **Functional Dependencies:**
 - *Review_ID* → *Order_ID*, *User_ID*, *Restaurant_ID*, *Rating*, *Comment*, *Created_At*
 - *Complaint_ID* → *Order_ID*, *User_ID*, *Description*, *Status*, *Created_At*

Proof: Determinants are Primary Keys. BCNF satisfied.

19. Cancellations & 20. Refunds

Cancellations(Cancellation_ID, *Order_ID*, Cancelled_By, Reason, Penalty_Amount, Created_At)

Refunds(Refund_ID, *Cancellation_ID*, Refund_Mode, Refund_Amount, Status, Completed_At)

- **Candidate Keys:** *Cancellation_ID*, *Order_ID* (1 cancellation per order), *Refund_ID*, *Cancellation_ID* (in Refunds table, 1 refund per cancellation)
- **Functional Dependencies:**
 - *Cancellation_ID* → *Order_ID*, *Cancelled_By*, *Reason*, *Penalty_Amount*, *Created_At*
 - *Refund_ID* → *Cancellation_ID*, *Refund_Mode*, *Refund_Amount*, *Status*, *Completed_At*

Proof: All determinants are candidate keys. BCNF satisfied.

21. Discounts

Discounts(Discount_ID, Code, Description, Percent, Max_Amount, Valid_From, Valid_Till)

- **Candidate Keys:** *Discount_ID*, *Code*
- **Functional Dependencies:**
 - *Discount_ID* → *Code*, *Description*, *Percent*, *Max_Amount*, *Valid_From*, *Valid_Till*
 - *Code* → *Discount_ID*, *Description*, *Percent*, *Max_Amount*, *Valid_From*, *Valid_Till*

Proof: Determinants are candidate keys. BCNF satisfied.

3. Cross-Check and Verification Checklist

This section verifies that all entities and relationships present in the provided ER Diagram (Quick-Bite_F12.pdf) and the Relational Schema (Quick-Bite_Relational_final_version_FF1.pdf) have been successfully accounted for and proven to be in BCNF.

Entity / Table Name	Present in ER Diagram	Covered in Proof	BCNF Verified?
Users	✓ Yes	✓ Yes	✓ Yes
Addresses	✓ Yes	✓ Yes	✓ Yes
Wallets	✓ Yes	✓ Yes	✓ Yes
Wallet_Transactions	✓ Yes	✓ Yes	✓ Yes
Restaurants	✓ Yes	✓ Yes	✓ Yes
Cuisines & Mapping	✓ Yes	✓ Yes	✓ Yes
Menu_Categories	✓ Yes	✓ Yes	✓ Yes
Menu_Items	✓ Yes	✓ Yes	✓ Yes
Carts	✓ Yes	✓ Yes	✓ Yes
Cart_Items	✓ Yes	✓ Yes	✓ Yes
Orders	✓ Yes	✓ Yes	✓ Yes
Order_Items	✓ Yes	✓ Yes	✓ Yes
Deliveries	✓ Yes	✓ Yes	✓ Yes
Delivery_Partners	✓ Yes	✓ Yes	✓ Yes
Payments	✓ Yes	✓ Yes	✓ Yes
Reviews	✓ Yes	✓ Yes	✓ Yes
Complaints	✓ Yes	✓ Yes	✓ Yes
Cancellations	✓ Yes	✓ Yes	✓ Yes
Refunds	✓ Yes	✓ Yes	✓ Yes
Discounts	✓ Yes	✓ Yes	✓ Yes

Conclusion: The exhaustive cross-check confirms that 100% of the tables derived from the Quick-Bite diagrams are documented. There are no partial dependencies (satisfying 2NF), no transitive dependencies of non-prime attributes (satisfying 3NF), and for every functional dependency $X \rightarrow Y$, X is a superkey. Therefore, the entire schema strictly adheres to the **Boyce-Codd Normal Form (BCNF)**.